

IZMIR INSTITUTE OF TECHNOLOGY

Department of Computer Engineering

CENG463 - Introduction to Machine Learning

Take-Home Midterm Examination Question 3

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1 Problem Formulation

The primary objective of this study is to conduct a comparative evaluation of linear, non-linear, and deep learning-based dimensionality reduction techniques using the MNIST dataset [2] [3]. High-dimensional data, such as the 28×28 pixel images of handwritten digits (784 dimensions), often contains significant redundancy and noise. Reducing this data to a lower-dimensional latent space (2D or 3D) is essential for visualization and feature extraction.

This analysis specifically focuses on how different mathematical approaches preserve the underlying structure of the data:

- **Linear Methods:** Evaluating PCA as a baseline to determine if simple orthogonal projections are sufficient to capture the variance of the digit classes.
- **Manifold Learning:** Implementing Kernel PCA (RBF), t-SNE, and UMAP to uncover the non-linear "manifolds" on which the digit data likely resides.
- **Deep Learning:** Designing an undercomplete autoencoder to learn a non-linear mapping through a bottleneck layer, prioritizing the minimization of reconstruction loss.

The core challenge is to balance the preservation of local structure (keeping similar digits together) and global structure (placing different digit clusters in a meaningful relative layout). We evaluate these models through a combination of:

- **Reconstruction Quality:** Measuring Mean Squared Error (MSE) to quantify information loss, particularly for PCA and the Autoencoder.
- **Structural Fidelity:** Using Trustworthiness and Kruskal's Stress to assess how well the low-dimensional embedding reflects the high-dimensional neighbor relationships.
- **Utility for Downstream Tasks:** Comparing the accuracy of a k-NN classifier ($k = 5$) trained on the reduced embeddings to determine which method produces the most discriminative features.

2 Methodological Decisions

The implementation strategy for this study was designed to systematically evaluate how different mathematical frameworks interpret the MNIST manifold. The following decisions were made for each algorithm:

- **Principal Component Analysis (PCA):** PCA was implemented as a linear baseline, reducing the 784-dimensional input to 2 components. This allows for a direct comparison of how much variance can be captured without non-linear mapping, which was found to be approximately 17.0%

- **Kernel PCA (RBF):** To account for the non-linear nature of handwritten digits, an **RBF kernel** was used with $\gamma = 0.01$. The `fit_inverse_transform=True` parameter was enabled to allow for the computation of Reconstruction MSE, ensuring a fair comparison with the Autoencoder and standard PCA.
- **t-SNE and Perplexity Tuning:** As required by the task, a grid search was conducted for perplexity values of 5, 30, and 50.

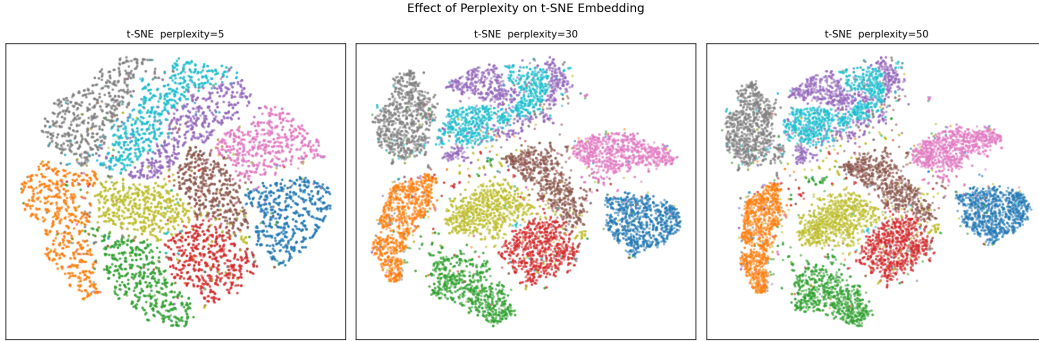


Figure 1: Effect of Perplexity on t-SNE Embeddings. Higher perplexity values demonstrate more coherent global structure and improved class separation.

As illustrated in Figure 1, lower perplexity (5) focuses heavily on local variations, while higher perplexity (50) produces more coherent and globally organized clusters, ultimately yielding the highest k-NN accuracy of 0.9447.

- **UMAP Parameter Exploration:** UMAP was tuned by varying `n_neighbors` (15, 30) and `min_dist` (0.1, 0.5).

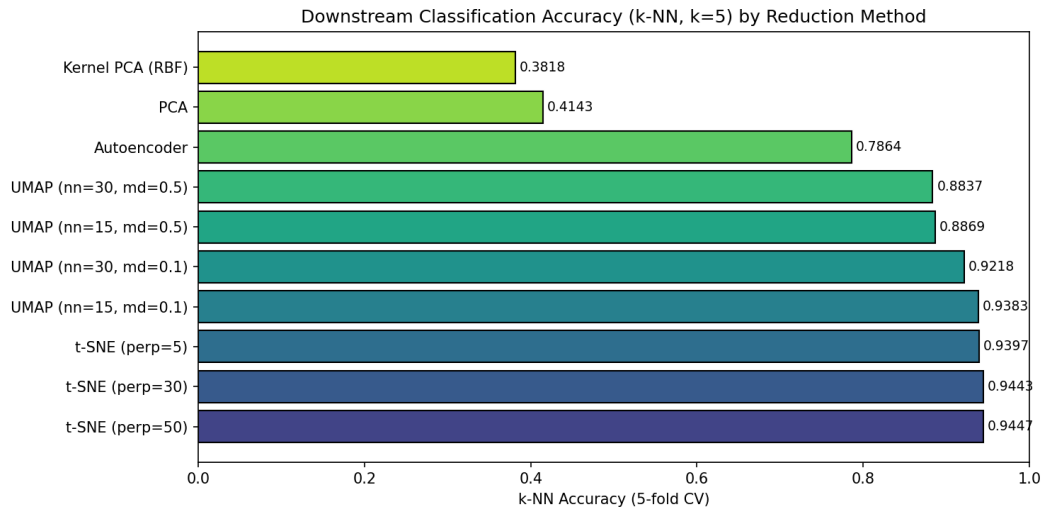


Figure 2: Comparison of Downstream k-NN Accuracy. UMAP and t-SNE variants significantly outperform linear methods.

The decision to prioritize a lower `min_dist` (0.1) was based on the need for tighter cluster formations, which significantly improved the downstream k-NN classification performance as seen in Figure 2.

- **Undercomplete Autoencoder Architecture:** The deep learning model was designed with a symmetric architecture:
 - **Encoder:** Three hidden layers with 512, 256, and 64 neurons, each followed by Batch Normalization and ReLU activation.
 - **Bottleneck:** A 2-neuron latent layer was chosen to facilitate direct 2D visualization.
 - **Optimization:** The model was trained for 50 epochs using the Adam optimizer and MSE loss.

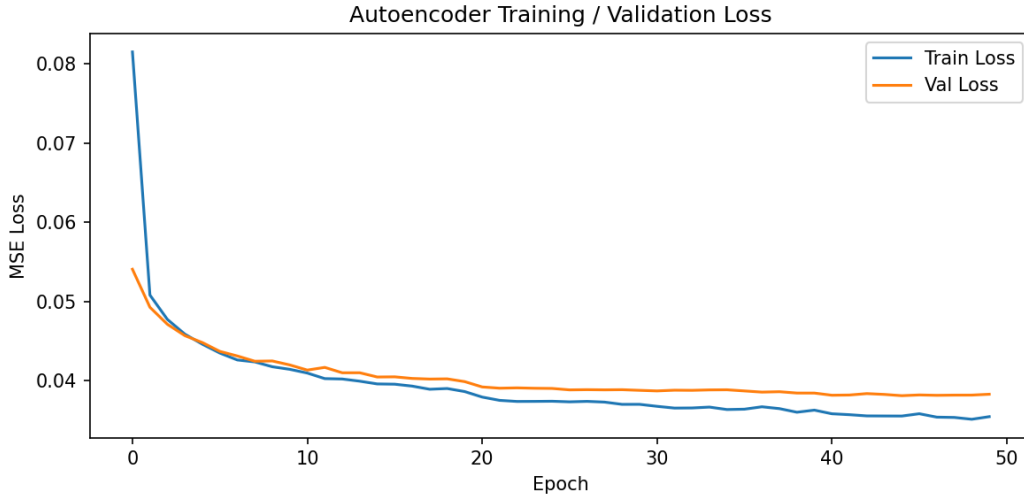


Figure 3: Autoencoder Training and Validation Loss over 50 Epochs. The convergence indicates a stable reconstruction state.

The loss curves in Figure 3 show smooth convergence, ensuring that the latent space features are well-learned before evaluation.

3 Experimental Setup

To ensure the reproducibility of the results and a fair comparison between models, the following experimental configurations were maintained:

- **Hardware and Software:** The experiments were conducted in a Google Colab environment utilizing an NVIDIA A100 GPU for accelerated deep learning training. The implementation relied on standard machine learning libraries, including `Scikit-Learn` for PCA and t-SNE, `umap-learn` for UMAP, and `PyTorch` for the Autoencoder implementation.

- **Dataset and Preprocessing:** The MNIST dataset (70,000 samples) was fetched using `OpenML`. Due to the high computational cost of manifold learning methods, a stratified subset of 10,000 samples was used for training and evaluation. Data was normalized by scaling pixel values to the range $[0, 1]$.
- **Reproducibility:** A global random seed of 22 was applied to all stochastic processes, including `NumPy` operations, `PyTorch` weight initializations, and the `train_test_split` generator.
- **Evaluation Pipeline:**
 - **Autoencoder:** An 80/20 split was used for training and validation to monitor convergence and prevent overfitting.
 - **Classification:** Downstream classification was performed using a k-NN classifier (k=5) with 5-fold stratified cross-validation on the reduced latent spaces to report mean accuracy and standard deviation.

4 Quantitative Evaluation

The performance of each dimensionality reduction technique was quantified using four primary metrics: Reconstruction Mean Squared Error (MSE), Trustworthiness, Kruskal’s Stress, and k-NN classification accuracy. The summarized results are presented in Table 1.

Table 1: Quantitative Comparison of Dimensionality Reduction Methods (n=10,000)

Method	MSE	Trust.	Stress	k-NN Acc.
PCA	0.0558	0.7329	N/A	0.4143 ± 0.0076
Kernel PCA (RBF)	0.0563	0.7208	N/A	0.3818 ± 0.0059
t-SNE (perp=50)	N/A	0.9906	5.6838	0.9447 ± 0.0060
UMAP (nn=15, md=0.1)	N/A	0.9612	0.4526	0.9383 ± 0.0070
Autoencoder	0.0350	0.9462	N/A	0.7864 ± 0.0079

4.1 Reconstruction Fidelity

As shown in Figure 4, the Autoencoder achieved the lowest reconstruction error (MSE = 0.0350), significantly outperforming the linear PCA baseline (MSE = 0.0558). This indicates that the non-linear deep layers successfully captured more complex image features than orthogonal projections, even with a bottleneck dimension of only 2. Interestingly, Kernel PCA with an RBF kernel showed a slightly higher MSE (0.0563) than standard PCA, suggesting that the chosen gamma value or the projection method was less efficient for reconstruction tasks on this specific subset.

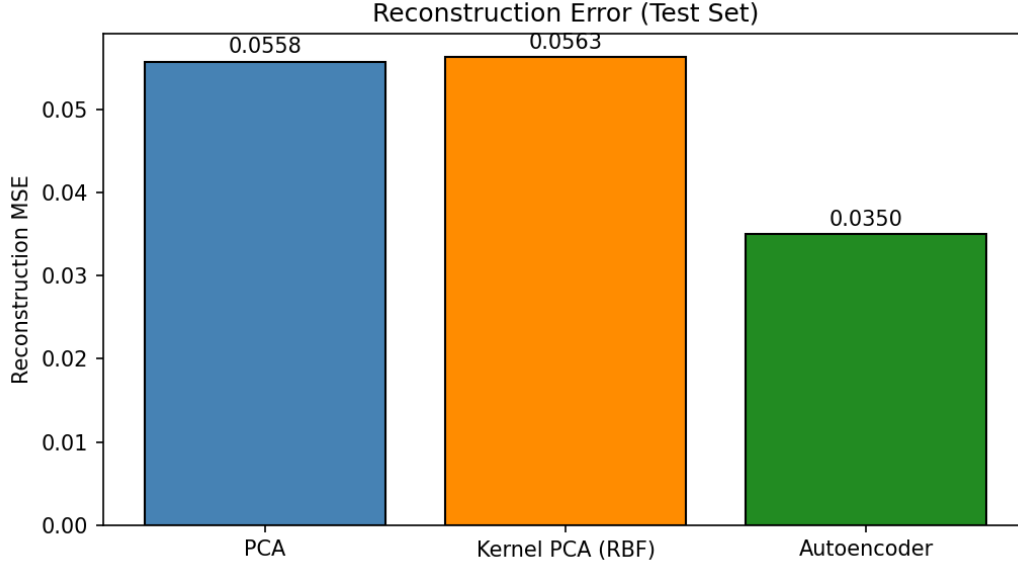


Figure 4: Reconstruction MSE comparison across PCA, Kernel PCA, and Autoencoder.

4.2 Manifold Preservation and Classification

The manifold learning methods (t-SNE and UMAP) demonstrated superior performance in preserving local structures and generating discriminative features .

- **Trustworthiness:** t-SNE (perp=50) achieved the highest trustworthiness score (0.9906), indicating that neighbor relationships in the 2D embedding are highly faithful to the original 784-D space.
- **Downstream Utility:** As visualized in Figure 2, t-SNE variants yielded the highest k-NN classification accuracies, peaking at 94.47%. While UMAP followed closely (93.83%), it demonstrated significantly lower Kruskal’s Stress (0.4526 vs 5.6838 for t-SNE), suggesting a better preservation of global distance relationships.
- **Autoencoder vs. Linear Methods:** The Autoencoder (78.64%) provided a massive improvement over PCA (41.43%) in classification utility, proving that deep learning can learn more representative latent features than traditional variance-maximization techniques.

5 Visual Analysis and Findings

5.1 Comparative Analysis of 2D Embeddings

As shown in Figure 5, there is a stark contrast between linear and manifold learning approaches in how they resolve the MNIST dataset.

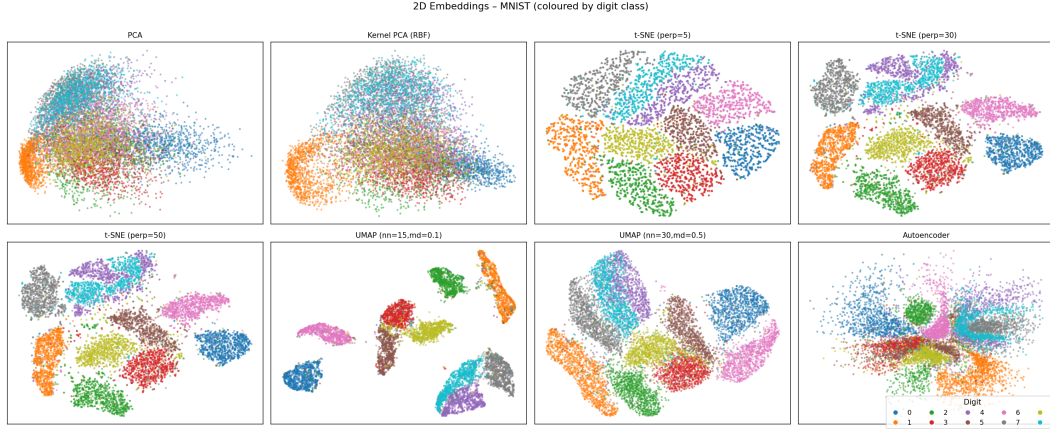


Figure 5: Complete comparison of 2D embeddings for all methods. Note the progression from high overlap in PCA to clear class separation in t-SNE and UMAP.

- **PCA and Kernel PCA:** Both methods exhibit significant overlap between classes, particularly for digits with similar topological structures such as 4, 7, and 9. While digit 1 (orange) and digit 0 (blue) are somewhat isolated, the central region remains a dense, inseparable cloud, confirming that linear projections fail to capture the nuances of handwritten strokes.
- **t-SNE:** The perplexity grid search shows that while **perp=5** creates small, fragmented clusters, **perp=50** generates well-defined "islands" for each digit. This alignment with high k-NN accuracy suggests that t-SNE is highly effective at preserving local neighborhood identities.
- **UMAP:** Similar to t-SNE, UMAP provides excellent separation. However, it tends to group similar digits (e.g., 3, 5, and 8) closer together in the global layout, reflecting its superior preservation of global manifold structure.
- **Autoencoder:** The Autoencoder creates a "continuous" map rather than isolated islands. While there is more overlap than in UMAP, the clusters follow a logical progression, with similar shapes appearing in adjacent regions.

5.2 Autoencoder Latent Space and Semantic Structure

A deeper look at the Autoencoder's performance in Figure 6 reveals how the model compresses information into two dimensions while maintaining reconstruction capability.

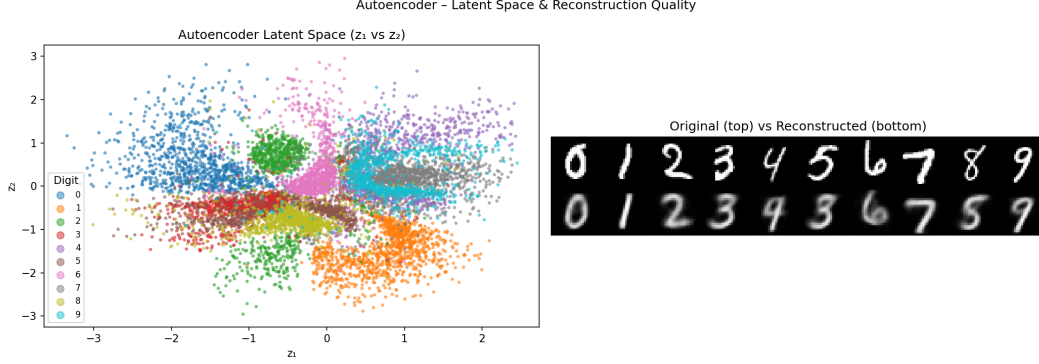


Figure 6: Left: Semantic distribution in the latent space. Right: Visual comparison of original digits versus their reconstructions from the 2D bottleneck.

The latent scatter plot (z_1 vs z_2) shows that the model has learned a coordinate system based on digit morphology. For instance, digit 1 (thin, vertical) is mapped to one extreme of the space, while digit 0 (round, hollow) occupies a distinct cluster on the opposite side.

Regarding the reconstruction quality shown in Figure 6, the reconstructed digits (bottom row) appear slightly blurred compared to the originals (top row). This "blurriness" is a direct result of the extreme compression (784 dimensions down to 2), which forces the model to prioritize the general "skeleton" of the digit over fine-grained pixel details. Despite this, every digit remains recognizable, demonstrating that the 2-unit bottleneck successfully captures the core semantic essence of the MNIST classes.

6 Discussion and Complexity Analysis

6.1 Computational Complexity

The execution times recorded in Table 2 highlight significant differences in the computational demands of each method:

Table 2: Quantitative Comparison of Dimensionality Reduction Methods (n=10,000)

Method	Time (s)	Recon MSE	Trust.	Stress	k-NN Acc.
PCA	0.18	0.0558	0.7329	—	0.4143 ± 0.0076
Kernel PCA (RBF)	7.09	0.0563	0.7208	—	0.3818 ± 0.0059
t-SNE (perp=5)	21.87	—	0.9925	8.7587	0.9397 ± 0.0050
t-SNE (perp=30)	27.27	—	0.9919	6.6294	0.9443 ± 0.0058
t-SNE (perp=50)	32.09	—	0.9906	5.6838	0.9447 ± 0.0060
UMAP (nn=15, md=0.1)	40.48	—	0.9612	0.4526	0.9383 ± 0.0070
UMAP (nn=15, md=0.5)	17.75	—	0.9342	0.4344	0.8869 ± 0.0046
UMAP (nn=30, md=0.1)	24.55	—	0.9538	0.4747	0.9218 ± 0.0065
UMAP (nn=30, md=0.5)	24.75	—	0.9365	0.4538	0.8837 ± 0.0066
Autoencoder	11.32	0.0350	0.9462	—	0.7864 ± 0.0079

- **PCA (0.18s):** The fastest method, utilizing efficient Singular Value Decomposition (SVD). Its complexity is roughly $O(\min(ND^2, N^2D))$, making it highly scalable for large datasets.
- **t-SNE (21s–32s):** Significantly more expensive due to the construction of a probability distribution over pairs of points. The Barnes-Hut implementation used here has a complexity of $O(N \log N)$, yet it remains slower than linear methods due to the iterative optimization process.
- **UMAP (17s–40s):** While theoretically faster than t-SNE ($O(N^{1.14})$), its execution time varied significantly based on hyperparameter configurations. It is generally preferred for large-scale manifold learning due to its scalability in higher dimensions.
- **Autoencoder (11.32s):** The training time is dependent on the number of epochs (50) and the architecture depth. While the initial training is slower than PCA, its inference time (transforming new data) is a single forward pass, making it faster than t-SNE for production environments.

6.2 Methodological Trade-offs

The choice of dimensionality reduction method involves a clear trade-off between preservation types:

- **Linear vs. Non-Linear:** PCA fails to resolve the MNIST dataset because handwritten digits exist on a non-linear manifold. As shown in Table 2, t-SNE and UMAP achieve >93% accuracy, more than doubling PCA’s performance (41.43%).
- **Local vs. Global Structure:** t-SNE (low Stress) excels at local clustering but can “tear” the global geometry. UMAP, with its significantly lower Kruskal’s Stress (0.4526), preserves the global relationship between clusters more effectively.
- **Parametric vs. Non-Parametric:** A key advantage of the Autoencoder and PCA is that they are parametric—they learn a mapping function ($f : X \rightarrow Z$) that can be applied to new data. In contrast, standard t-SNE is non-parametric and would require re-running the entire optimization if new samples were added to the dataset.

7 Conclusion

This study demonstrates that while linear methods like PCA provide a fast baseline and reasonable reconstruction ($\text{MSE} = 0.0558$), they are insufficient for tasks requiring high class discriminability on image data. Manifold learning techniques, specifically t-SNE with a perplexity of 50, yielded the best feature separation for classification (94.47% accuracy).

However, the Undercomplete Autoencoder emerged as the most versatile tool, achieving the best reconstruction fidelity ($\text{MSE} = 0.0350$) and providing a balanced middle ground for classification. For real-world applications where both visualization and the ability to process new data are required, the Autoencoder or UMAP would be the superior choices over the computationally heavy and non-parametric t-SNE.

Repository and Reproducibility

GitHub Link: <https://github.com/behicekadioglu/CENG463---Take-Home-Midterm/tree/main>

References

- [1] Prof. Dr. Aytuğ Onan, *CENG 463 Machine Learning Take Home Midterm*, Izmir Institute of Technology, 2026.
- [2] Feurer, M., van Rijn, J. N., Kadra, A., et al. (2021). *OpenML-Python: an extensible Python API for OpenML*. Journal of Machine Learning Research, 22(100), 1-5.
- [3] Lecun, Y., Bottou, L., Bengio, Y., and Haffner, P. (1998). *Gradient-based learning applied to document recognition*. Proceedings of the IEEE, 86(11), 2278-2324.